CS-523 Course Project Proposal

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Project name

Simple Workflow Simulator

Topic area

A simulation of a simple DAG-based workflow system

Project vision

This project offers a simple Directed Acyclic Graph (DAG) based workflow interface and its execution simulation.

The Workflow is defined as and limited to DAG, composed of one or more Tasks and their dependencies.
Each Task has some simple parameters like this:

- Duration of execution time
- Dependencies
- Failure rate
- Allowed retry count

The Workflow is defined in the custom format such as a toml or yaml file.

When executed, the program shows the status and progress of workflow through a simple TUI.
A rough mock-up is shown below:

Worker	Task	Progress	
Worker A	Task1	▓▓▓▓▒▒▒▒▒▒	50%
Worker B	Task2	▓▓▓▓▓▓▒▒▒▒	70%
Worker C	Task4	▓▒▒▒▒▒▒▒▒▒	20%

Elapsed time: 50 seconds

Emergent behavior

Tasks follow some simple rules:

- Have a status such as Waiting, Ready, Running, Completed
- Its status becomes Ready when all of its dependencies are completed.
- Workers start executing ready tasks when they become idle
- Tasks may be retried after failure if retry limits remain

From these simple rules, I hope to observe a larger-scale, whole workflow behavior and differences in completion time under different parameters and workflow structures.

Initial goals

- Define workflow format
- Be able to simulate the execution of the defined workflow
- Display the progress of workflow on a simple TUI

Stretch goals

- Consider resource consumption. That is, simulating the CPU and Memory limits and usage
- Be able to switch scheduling policy that determines which worker each task should be assigned to
- Rich visualization of drawing workflow graph

Issues of concern

- To observe Emergent Behavior, it would be better to use a reasonably large workflow for simulation. However, it's tedious to write that workflow definition from scratch, so I want to set up a kind of default workflow template or parameters. The concern is how to generate or provide such workflows efficiently.

Git repo URL

https://github.com/harutofujihara/simple-workflow-simulator

AI Acknowledgement

I discussed with ChatGPT how to adapt my idea to the project theme. ChatGPT also helped correct a few parts of my English that were clearly incorrect or confusing.